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wPINNs: Enhancing Physics-Informed Neural Networks with Wavelet Activation Functions

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서울대학교 자연과학대학
물리·천문학부 (물리학전공)
2019-11712
주다현

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이 논문을 이학학사 학위 논문으로 제출함

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wPINNs: Enhancing Physics-Informed Neural Networks with Wavelet Activation Functions

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Physics-Informed Neural Networks (PINNs) have emerged as versatile tools for solving PDE-driven problems by embedding physical laws into the training framework. However, standard PINNs, relying on MLPs with hyperbolic tangent (tanh) activation functions, often fall short in modeling temporal dependencies. PINNsFormer, a Transformer-based extension of PINNs, addresses these limitations with multi-head attention mechanisms and a Wavelet activation function but incurs significant computational overhead. This study introduces wPINNs, which integrates Wavelet activation into the conventional PINN framework to enhance accuracy without substantial computational cost. Experiments on the Karman vortex problem demonstrate that wPINNs drastically lower error compared to PINNs while preserving computational efficiency. Robustness tests highlight logical outlier handling in wPINNs, while PINNs exhibit unexpected behaviors with anomalies, warranting further investigation. The results confirm the utility of Wavelet activations for advancing PINNs, providing a practical solution for computational physics challenges.

keywords : wPINNs, PINNs, PINNsFormer, Wavelet activation function, Navier-Stokes equations, Karman vortex street
Student Number: 2019-11712

1. Introduction

Physics-Informed Neural Networks (PINNs) have gained significant traction in solving forward and inverse problems related to nonlinear partial differential equations (PDEs). By embedding physical laws into the training process, PINNs leverage observational data and governing equations to approximate solutions while satisfying PDE constraints (Raissi et al., 2019). This unique approach eliminates the need for large-scale labeled data, making PINNs particularly useful in computational physics and engineering.

Despite their success, conventional PINNs employ multilayer perceptrons (MLPs) with hyperbolic tangent (tanh) activation functions, which do not account for temporal dependencies in physical systems. This limitation leads to challenges in modeling complex dynamical phenomena and accurately capturing solutions for time-dependent PDEs. Furthermore, while PINNs show robustness in many scenarios, their performance under conditions involving data anomalies or outliers remains a point of contention.

To address these challenges, Zhao et al. (2024) introduced PINNsFormer, a Transformer-based framework that incorporates temporal dependencies using attention mechanisms and employs the Wavelet activation function. PINNsFormer demonstrated superior accuracy but at the cost of significantly increased computational time, creating a bottleneck for practical applications.

This study aims to address the following key questions:

1. Performance Enhancement with Minimal Cost: Can the Wavelet activation function, used in PINNsFormer, improve the performance of conventional PINNs without significantly increasing computational time?
2. Understanding Robustness: Is the robustness of PINNsFormer superior to conventional PINNs under scenarios involving manipulated (outlier) data, and how does wPINNs (Wavelet-enhanced PINNs) compare in this regard?

To answer these questions, the author introduces wPINNs, a modified PINNs architecture that integrates the Wavelet activation function. The author evaluates the performance of PINNs, PINNsFormer, and wPINNs on the Navier-Stokes equation with and without data manipulation to assess computational efficiency, accuracy, and robustness.

In this paper, the term "relative L1 error" refers to the summation of absolute errors (Total Sum of Errors, TSE) across all data points, providing a clear metric for evaluating the accuracy of predictions. Additionally, the words "manipulated" and "measured" are used interchangeably to describe data that contains outliers or errors, while the terms "real," "standard," and "original" are used synonymously to refer to the true, unaltered data.

2. Background

2.1 Physics-Informed Neural Networks (PINNs)

Physics-Informed Neural Networks (PINNs) are deep learning models designed to solve partial differential equations (PDEs) by embedding the governing physical laws directly into the training process (Raissi et al., 2019). The loss function in PINNs combines two main components: the data loss, which measures the discrepancy between observed and predicted data, and the physics loss, which ensures that the predictions satisfy the PDE constraints. This enables PINNs to use spatial and temporal inputs to predict solution variables, such as velocity and pressure in fluid dynamics.

However, conventional PINNs rely on multilayer perceptrons (MLPs) with hyperbolic tangent (tanh) activation functions and a point-to-point prediction framework. These models lack the ability to explicitly account for temporal dependencies, which are critical in physical systems where causality and dynamic interactions play a fundamental role. As a result, the performance of PINNs diminishes in problems requiring a global understanding of system dynamics, such as those involving long-term temporal dependencies.

2.2 PINNsFormer and the Wavelet Activation Function

To address these limitations, Zhao et al. (2024) proposed PINNsFormer, a Transformer-based framework that integrates temporal dependencies into the PINNs architecture. PINNsFormer achieves this by transforming point-wise inputs into pseudo-sequences and employing a multi-head attention mechanism, as illustrated in Figure 1. This design enables PINNsFormer to propagate initial condition constraints globally, ensuring more accurate solutions for complex, time-dependent PDE scenarios.

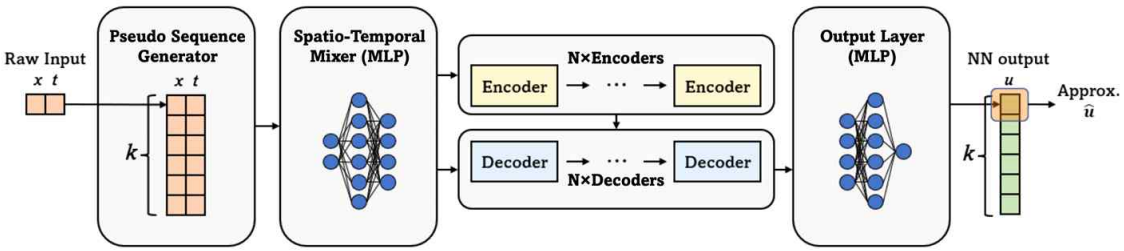


Fig. 1 Architecture of proposed PINNsFormer. PINNsFormer generates a pseudo sequence based on pointwise input features. It outputs the corresponding sequential approximated solution. The first approximation of the sequence is the desired solution u_hat(x,t).(Zhao et al., 2024)[2]

Another key feature of PINNsFormer is the introduction of the Wavelet activation function, defined as:

$$Wavelet(x) = \omega_1 \sin(x) + \omega_2 \cos(x) \qquad (1)$$

where ω_1 and ω_2 are learnable parameters (Zhao et al., 2024). This activation function leverages the mathematical properties of Fourier transforms to approximate periodic and oscillatory behavior, making it particularly effective for physical systems characterized by such dynamics. In contrast to the tanh activation function, Wavelet offers greater flexibility and approximation power for solutions involving oscillatory or multiscale features, which are common in PDEs.

Moreover, the encoder-decoder architecture of PINNsFormer (depicted in Figure 2) incorporates multi-head attention in the encoder to capture long-term dependencies across all spatiotemporal input features. However, the decoder omits self-attention layers to focus on the specific temporal predictions required for solving PDEs. This architecture ensures a robust

and dynamic modeling of time-dependent physical systems.

The advantages of PINNsFormer are further illustrated through a comparison of its loss landscape with that of conventional PINNs, as shown in Figure 3. The loss landscape for PINNsFormer is significantly smoother, with fewer local minima and sharp cones, enabling more stable optimization and convergence during training. This smoother landscape, coupled with the Wavelet activation function, results in superior accuracy and generalization across various PDE problems.

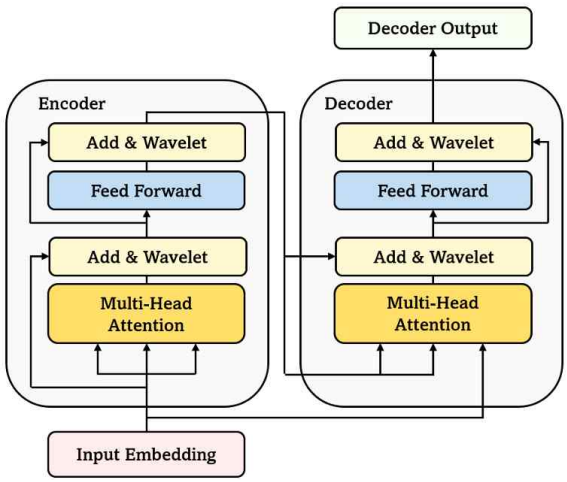


Fig. 2 The architecture of PINNsFormer's Encoder-Decoder Layers. The decoder is not equipped with self attentions.(Zhao et al., 2024) [2]

Despite its advantages, PINNsFormer incurs a significant computational cost due to the complexity of the Transformer-based architecture. This motivates the exploration of simpler architectures, such as conventional PINNs enhanced with the Wavelet activation function, to achieve a balance between accuracy, robustness, and computational efficiency.

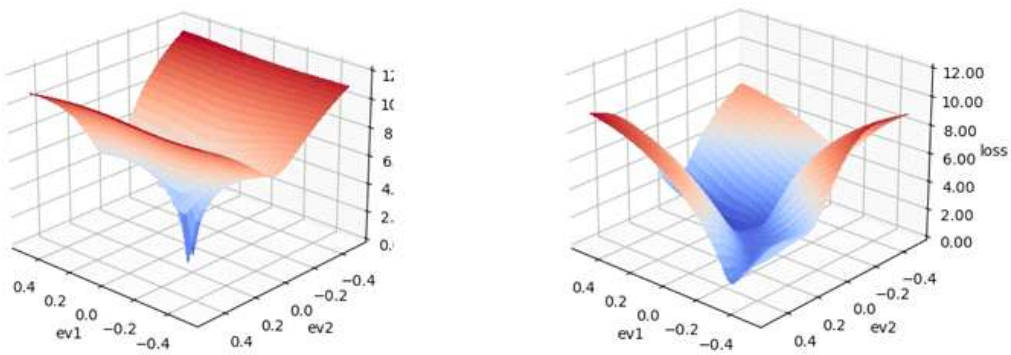


Fig. 3 Visualization of the loss landscape for PINNs (left) and PINNsFormer (right) on a logarithmic scale. The loss landscape of PINNsFormer is significantly smoother than conventional PINNs.(Zhao et al., 2024) [2]

3. Methodology

3.1 Problem Definition

This study evaluates the influence of activation functions on the performance and robustness of PINNs when handling outlier data. To achieve this, the study focuses on solving the Navier-Stokes equations for incompressible fluid flow around a circular cylinder, a well-studied problem in computational fluid dynamics (Raissi et al., 2019). This scenario is chosen because of its rich, dynamic nature, which involves periodic vortex shedding, making it a challenging benchmark for PINNs.

3.2 Navier-Stokes Equations

The Navier-Stokes equations describe the motion of incompressible fluid substances and are fundamental in fluid dynamics. For a two-dimensional, incompressible flow, the Navier-Stokes equations consist of the following components:

- Momentum equations:

$$\frac{\partial u}{\partial t}+u\frac{\partial u}{\partial x}+v\frac{\partial u}{\partial y}=-\frac{\partial P}{\partial x}+\nu\left(\frac{\partial^2 u}{\partial x^2}+\frac{\partial^2 u}{\partial y^2}\right) \qquad (2)$$

$$\frac{\partial v}{\partial t}+u\frac{\partial v}{\partial x}+v\frac{\partial v}{\partial y}=-\frac{\partial P}{\partial y}+\nu\left(\frac{\partial^2 v}{\partial x^2}+\frac{\partial^2 v}{\partial y^2}\right) \qquad (3)$$

- Continuity equation:

$$\frac{\partial u}{\partial x}+\frac{\partial v}{\partial y}=0 \qquad (4)$$

Here, u and v are the velocity components in the x- and y-directions, p is the pressure, and ν is the kinematic viscosity. These equations model the time-dependent flow of an incompressible fluid.

3.3 Data Generation and Manipulation

The flow past a circular cylinder is simulated using spectral methods, following the setup outlined by Raissi et al. (2019). This simulation generates high-resolution spatiotemporal data for velocity components and pressure fields. The training data corresponds to the wake region behind

the cylinder, capturing the dynamic vortex shedding phenomenon known as the Karman vortex street, a canonical pattern in fluid dynamics. Figure 4 visualizes the computational domain, including the vortex shedding and the training data points.

To test robustness, a deliberate data manipulation is introduced by altering the pressure field. Specifically, the 2150th data point in the training set is assigned an abnormally large value (−1), simulating a measurement error or experimental anomaly. This outlier represents a significant deviation from surrounding data and challenges the model's ability to learn meaningful physical relationships.

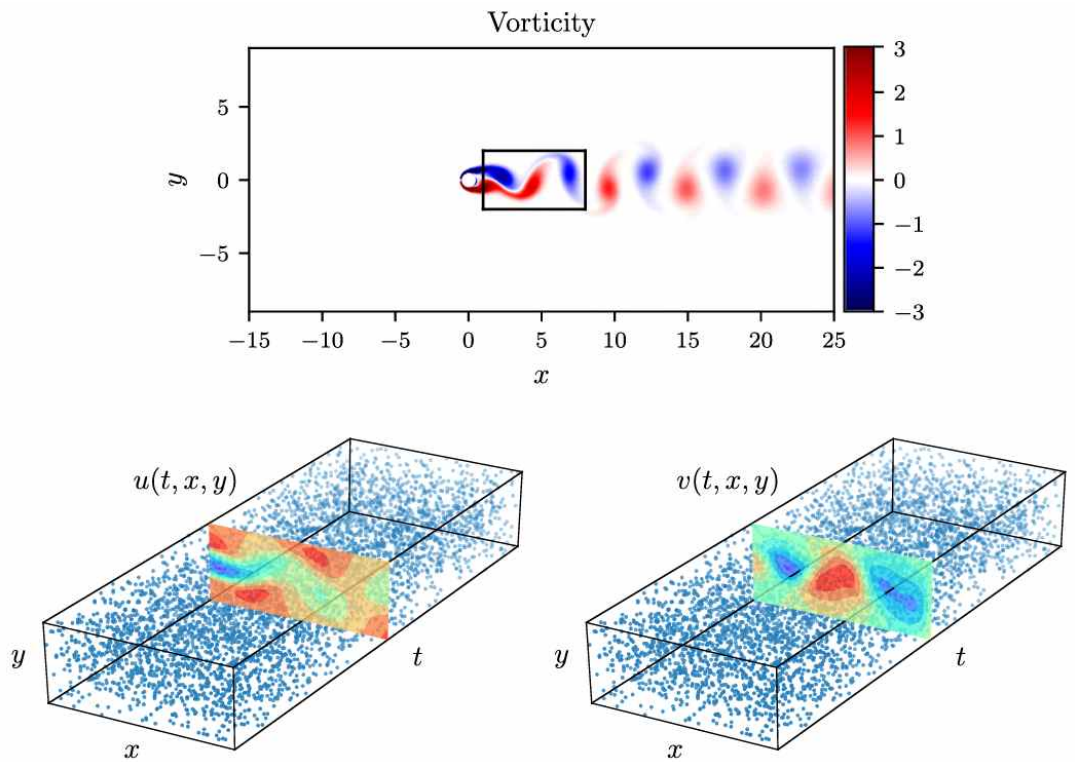


Fig. 4 *Navier-Stokes equations*: Top: Incompressible flow and dynamic vortex shedding past a circular cylinder at Re =100. The spatio−temporal training data correspond to the depicted rectangular region in the cylinder wake. Bottom: Locations of training data−points for thestream−wise and transverse velocity components, u(t, x, y)and v(t, x, t), respectively.(Raissi et al., 2019)[1]

3.4 Model Architectures

- Conventional PINNs with Tanh Activation

The conventional PINNs architecture is implemented as described by

Raissi et al. (2019). This model uses a multilayer perceptron (MLP) with tanh activation functions in its hidden layers. The inputs to the network are the spatiotemporal coordinates (x,y,t), and the outputs are the predicted velocity components (u,v) and pressure (p). The loss function is a weighted sum of:

1. Data loss: Quantifies the difference between predicted and observed values at training points.
2. Physics-informed loss: Enforces the residuals of the Navier-Stokes equations and the continuity equation to approach zero.

- Modified PINNs with Wavelet Activation

The author modifies the conventional PINNs by replacing the tanh activation function with the Wavelet activation function (Equation 1) introduced in PINNsFormer (Zhao et al., 2024). The network architecture, including the number of layers and neurons, remains identical to the conventional PINNs, except for the activation function used in the hidden layers.

The Wavelet activation function's design, inspired by Fourier transforms, allows the network to approximate periodic and oscillatory behaviors effectively, making it a promising alternative for improving PINNs' performance in dynamic physical systems.

- Training Parameters

Both models—conventional PINNs and modified PINNs with Wavelet activation (referred to as wPINNs)—are trained using the L-BFGS optimizer, a quasi-Newton method known for its effectiveness in handling small to medium-sized datasets (Raissi et al., 2019). The training involves minimizing a combined loss function that includes data and physics-informed losses, ensuring a fair comparison. The computational time for training is recorded to evaluate efficiency.

- Evaluation Metrics

The models' performance is assessed using the Total Sum of Errors (TSE), which measures the discrepancy between the predicted and true

pressure fields:

$$TSE = \sum_i |p_{pred}(x_i, y_i, t_i) - p_{true}(x_i, y_i, t_i)| \qquad (5)$$

where p_{pred} is the predicted pressure, p_{true} is the ground truth, and the sum is over all data points in the test set.

Additionally, the models are evaluated qualitatively by visualizing the predicted pressure fields and examining their alignment with the expected flow characteristics. The models' robustness is tested using both the clean training dataset and the manipulated dataset containing outliers.

4. Results

4.1 Validation of PINNsFormer Robustness

PINNsFormer exhibited exceptional robustness to single outlier data (Figure 5). The training time for PINNsFormer remained nearly constant—approximately 5837.8283 seconds for the clean dataset (Figure 6) and 5840.6867 seconds for the dataset with one manipulated outlier. The relative L1 error showed negligible change, with 3.835758 for clean data and 3.816996 for manipulated data. This consistency confirms PINNsFormer's stability when exposed to anomalous data, attributed to its Transformer-based architecture and attention mechanisms that capture temporal dependencies across the dataset.

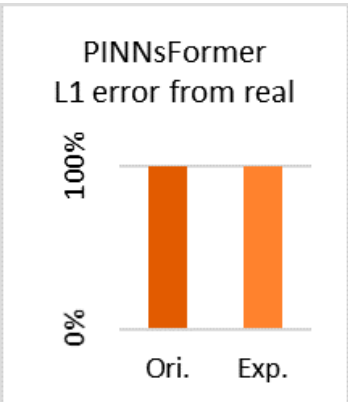


Fig. 5 PINNsFormer error results conducted and compared with the real data.

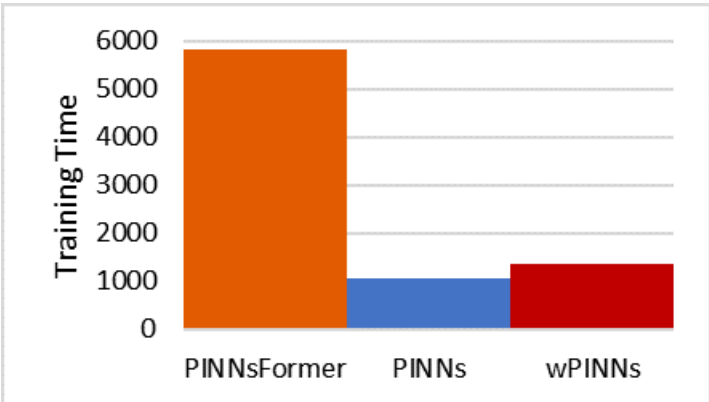


Fig. 6 PINNsForemr, PINNs, and wPINNs training time. The results were compared with the results conducted by the real data.

4.2 Accuracy Gains with Wavelet Activation in wPINNs

wPINNs demonstrated substantial improvements in accuracy over conventional PINNs for clean data. While PINNs required only 1057.998 seconds for training, its relative L1 error was a high 4.937524, highlighting its limited ability to approximate the solution. In contrast, wPINNs, with a modest increase in training time to 1373.7776 seconds, achieved a relative L1 error of 0.195548 (the training time data are shown in Figure 6, and the errors are depicted in Appendix B figures).

This significant reduction in error emphasizes the effectiveness of the Wavelet activation function, which enables better representation of periodic and oscillatory features inherent in PDE solutions without incurring excessive computational overhead.

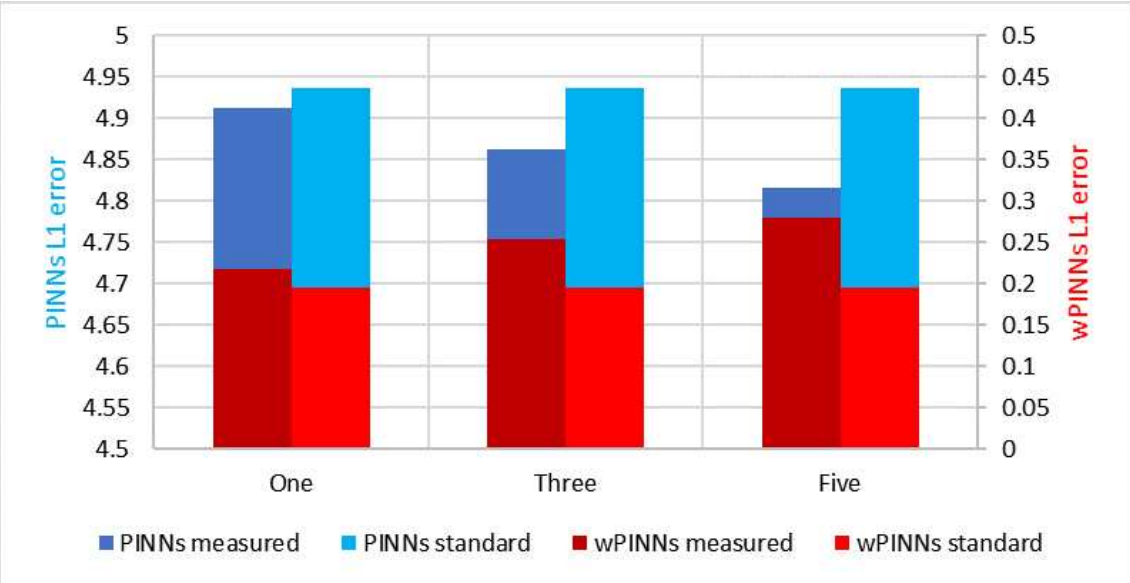


Fig. 7 Error comparison for PINNs and wPINNs with manipulated datasets containing one, three, and five outliers (all −1). The results highlight that wPINNs maintain consistent accuracy for the clean dataset while demonstrating an increase in error with additional outliers, whereas PINNs exhibit an unexpected decrease in error with additional outliers.

4.3 Balancing Computational Efficiency and Accuracy

The training times for PINNs, wPINNs, and PINNsFormer are compared in Figure 7, highlighting the computational efficiency of each model. PINNsFormer delivered the highest accuracy but at the cost of a disproportionately long training time of over 5837 seconds.

In contrast, wPINNs offered a middle ground. It improved accuracy significantly while increasing training time moderately, making it a more practical alternative for many applications. The balance achieved by wPINNs suggests that the Wavelet activation function is a computationally efficient enhancement compared to the computationally demanding Transformer-based PINNsFormer.

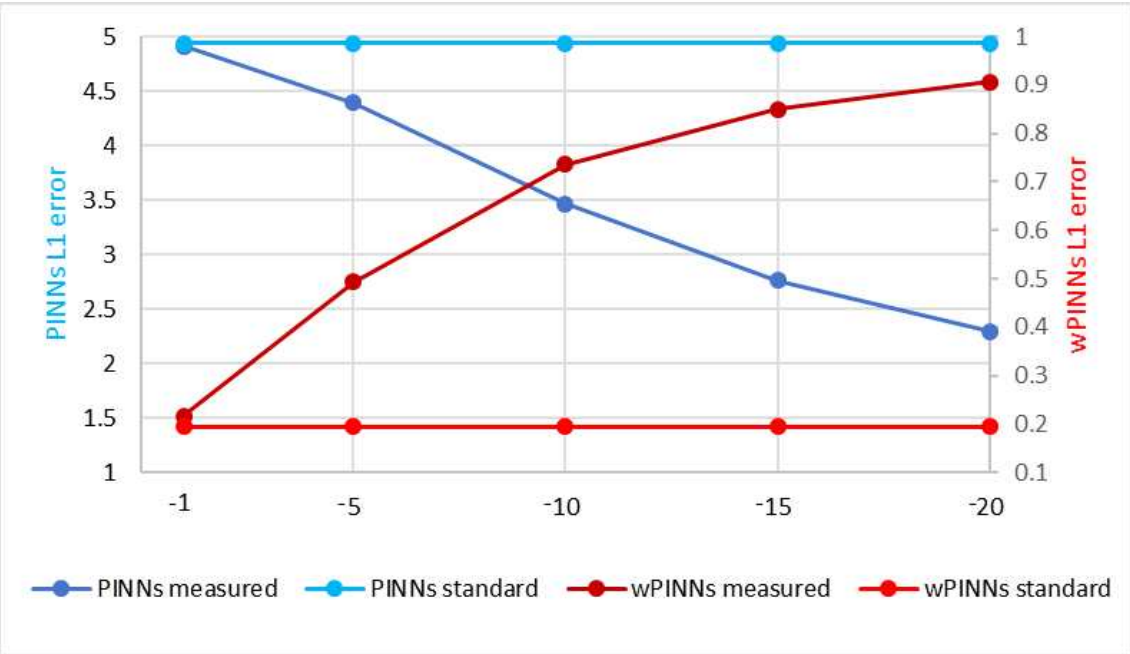


Fig. 8 Error trends for PINNs and wPINNs under manipulated datasets with varying levels of outlier severity. While PINNs exhibit unexpected decreases in error under extreme outliers, wPINNs show expected increases in error as outlier severity rises, maintaining robustness and preserving accuracy for the clean dataset.

5. Discussion

5.1 Influence of Activation Functions on Robustness and Accuracy

The Wavelet activation function's sinusoidal basis proved crucial in enhancing wPINNs' accuracy. By leveraging properties akin to Fourier transforms, the Wavelet function can effectively capture the oscillatory behaviors characteristic of PDE solutions. This alignment enables wPINNs to resolve finer details in the data, particularly periodic features, that the

tanh activation function used in PINNs fails to approximate.

The robustness of wPINNs under manipulated data further validates its efficacy. As shown in Figure 7 and 8, wPINNs maintained stable error levels for clean data, even as the severity or quantity of outliers increased. This robustness stems from the Wavelet function's ability to isolate and minimize the influence of anomalies without disrupting baseline performance.

However, an unexpected trend was observed in PINNs: errors decreased as outlier severity or quantity increased (Figure 7 and 8). This unexpected phenomenon suggests potential artifacts in the optimization process. One hypothesis is that the already large baseline errors in PINNs may overshadow the added impact of outliers, rendering the additional noise less relevant to the overall loss minimization. This counterintuitive behavior warrants further investigation to uncover the underlying dynamics of optimization in noisy datasets.

5.2 Practical Implications for Neural Network Design in Physics

The results highlight the critical role of activation functions in the performance and robustness of neural networks designed for physics-informed tasks. The Wavelet activation function, by bridging the gap between computational efficiency and accuracy, emerges as a strong candidate for enhancing the performance of PINNs without the heavy computational demands of Transformer-based models like PINNsFormer.

The ability of wPINNs to achieve near-PINNsFormer levels of accuracy while maintaining computational efficiency makes it a practical choice for real-world applications. Moreover, the Wavelet function's alignment with the spectral properties of physical systems ensures a more interpretable and reliable model, which is crucial for computational physics.

These findings pave the way for hybrid architectures that combine the computational efficiency of wPINNs with select features of PINNsFormer, such as attention mechanisms, to further optimize scalability and accuracy.

The Wavelet activation function could also be explored in other physics-informed machine learning frameworks to address similar challenges.

6. Conclusion

This study highlights the effectiveness of the Wavelet activation function in enhancing the accuracy and robustness of wPINNs while maintaining computational efficiency. PINNsFormer demonstrated notable stability under outlier conditions, showing minimal changes in error and training time, thereby validating its robustness. By integrating the Wavelet activation function, wPINNs significantly improved the accuracy of solutions compared to conventional PINNs, with only a modest increase in training time. This showcases the potential of the Wavelet activation function as a practical enhancement to PINNs, striking a balance between performance and computational demands without the extensive overhead associated with models like PINNsFormer.

Future work should investigate the unexpected phenomenon observed in PINNs, where errors decreased under certain outlier conditions, and examine its implications for robustness and optimization processes. Additionally, exploring the integration of the Wavelet activation function into advanced architectures like PINNsFormer could further improve both scalability and precision. A deeper theoretical analysis of the Wavelet function's ability to approximate PDE solutions could provide valuable insights into its efficacy. By aligning neural network design with the intrinsic characteristics of physical systems, researchers can develop more robust and efficient tools for solving complex computational physics problems.

7. Appendix

The simulation results are structured to provide a comprehensive view of the data and model performance. In each figure, the manipulated (or

measured) data, which includes any introduced outliers, is displayed in the top-left plot, while the real (original or standard) data, serving as the benchmark, is shown in the bottom-left plot. The model' s simulation result, representing its predicted output, is depicted in the top-right plot. Lastly, the absolute error (relative L1 error) between the simulation result and the real data is illustrated in the bottom-right plot, providing a clear visualization of the model's accuracy and deviations from the ground truth. This arrangement allows for an intuitive and detailed comparison of the data and results.

For the simulation conducted with the real data, the top-left plot and the bottom-left plot are identical, as the measured data in this case is the same as the real (original or standard) data. This indicates that no outliers or anomalies were introduced into the dataset, providing a baseline for evaluating the model's performance under normal conditions.

Appendix A. PINNsFormer robustness validation

PINNsFormer

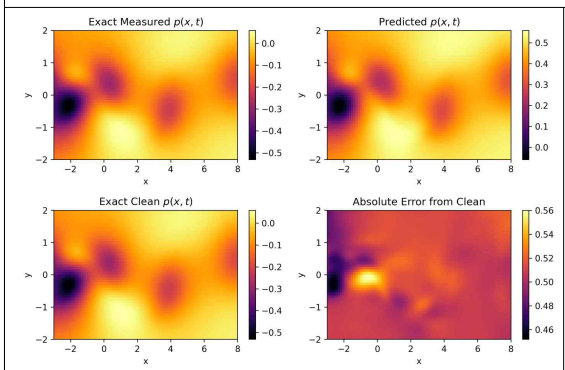


Fig. 9 PINNsFormer original

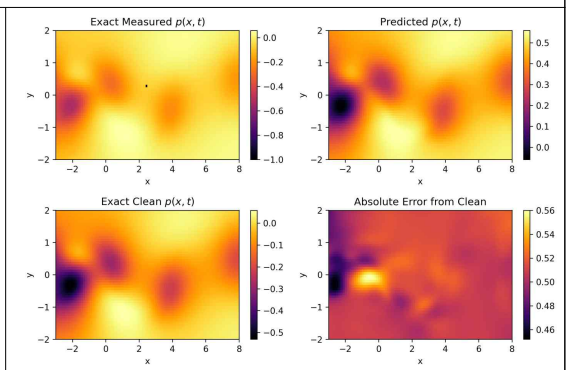


Fig. 10 PINNsFormer manipulated

Appendix B. PINNs and wPINNs performance comparison

PINNs	wPINNs

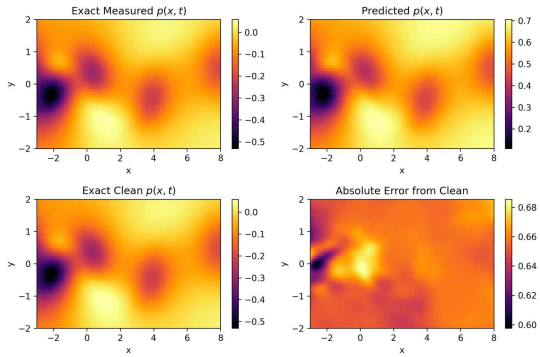
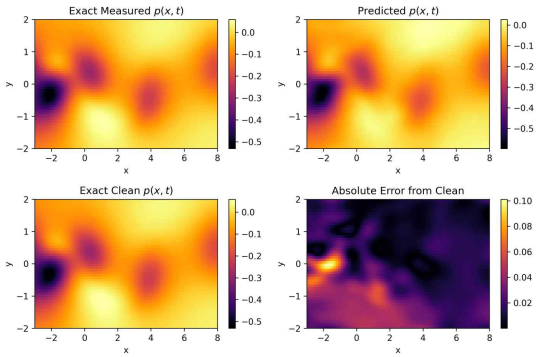

	

Fig. 11 PINNs original	**Fig. 12** wPINNs original

Appendix C. Making the outlier to deviate more from the real value

PINNs	wPINNs

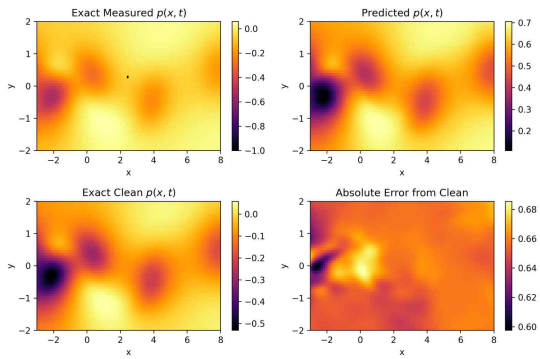
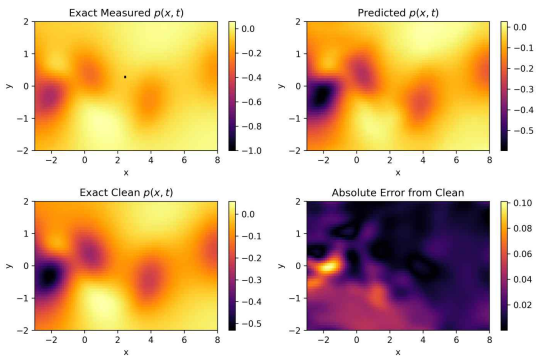

	

Fig. 13 PINNs 2150th −1 manipulated	**Fig. 14** wPINNs 2150th −1 manipulated

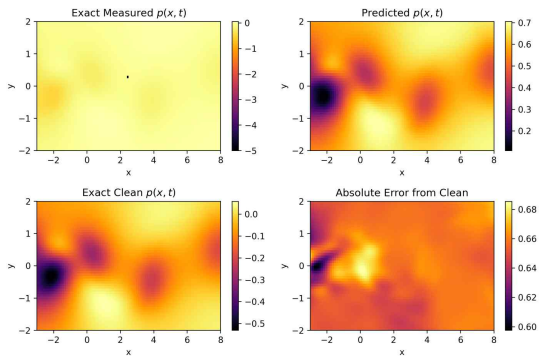
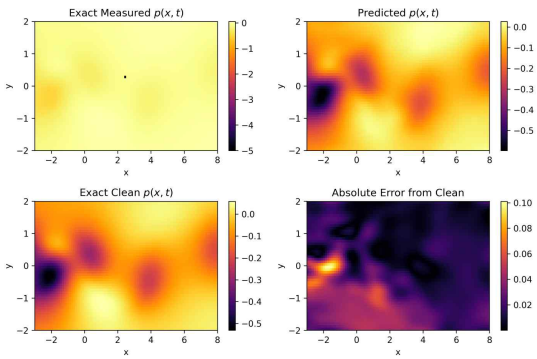

	

Fig. 15 PINNs 2150th −5 manipulated	**Fig. 16** wPINNs 2150th −5 manipulated

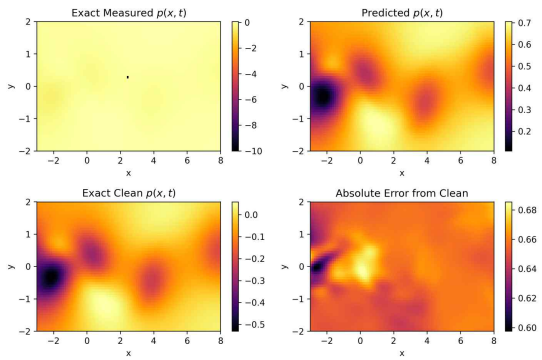


Fig. 17 PINNs 2150th −10 manipulated

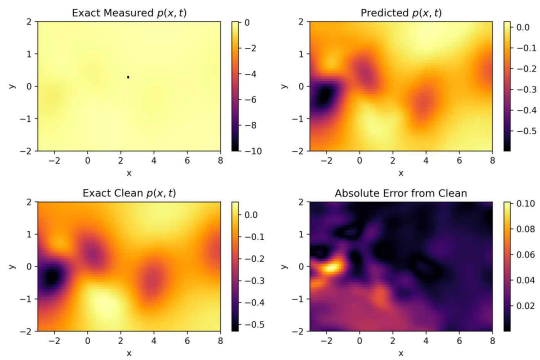


Fig. 18 wPINNs 2150th −10 manipulated

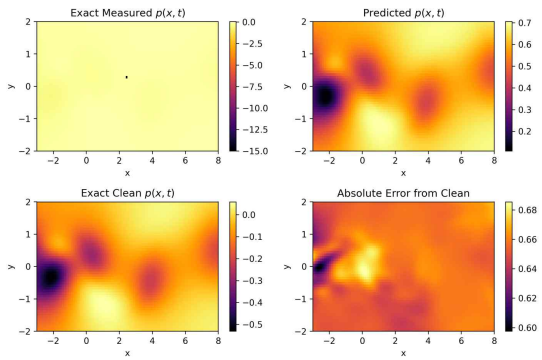


Fig. 19 PINNs 2150th −15 manipulated

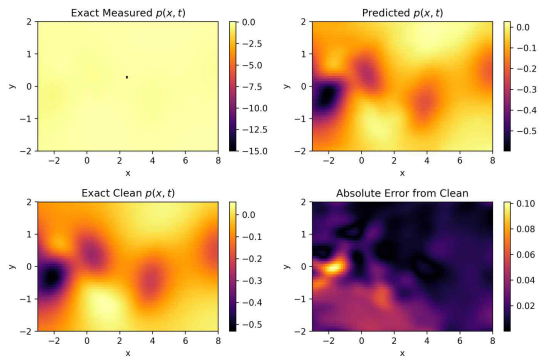


Fig. 20 wPINNs 2150th −15 manipulated

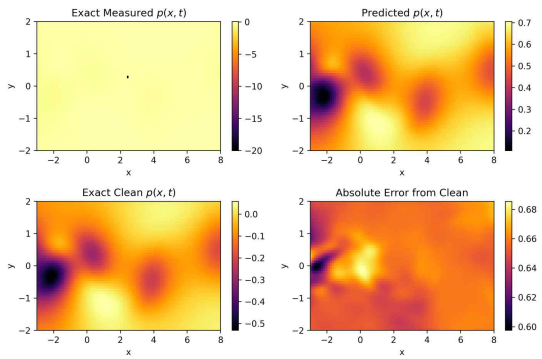


Fig. 21 PINNs 2150th −20 manipulated

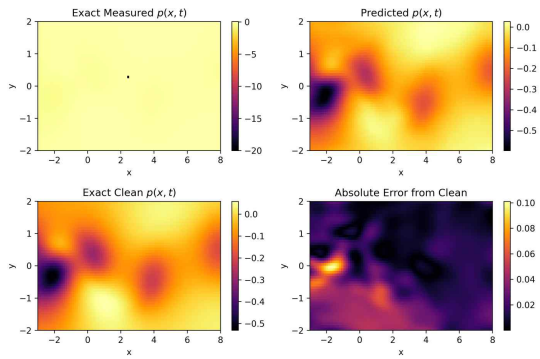
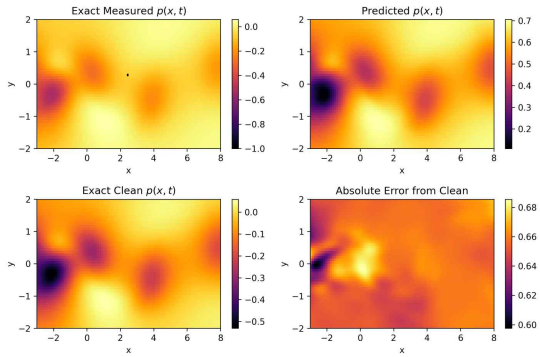
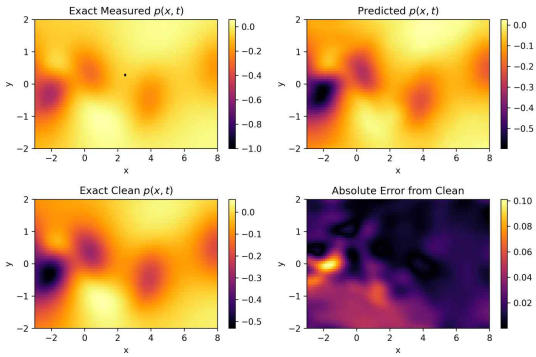
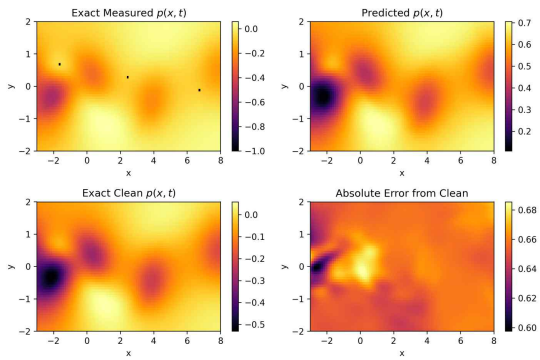
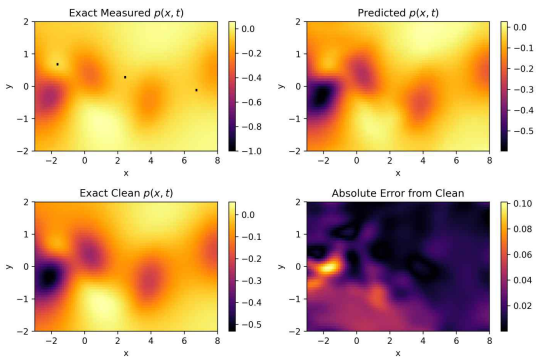
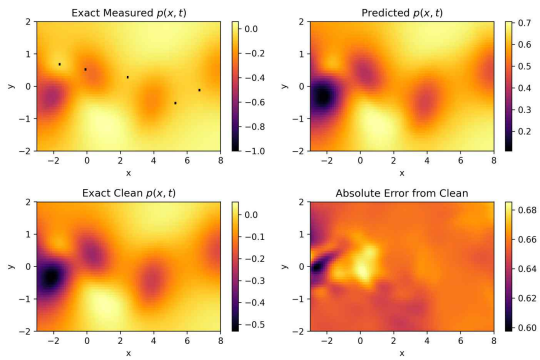
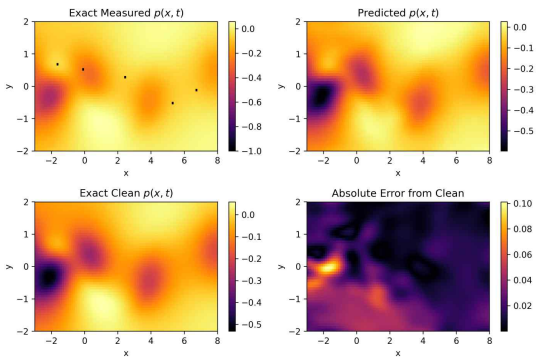


Fig. 22 wPINNs 2150th −20 manipulated

Appendix D. Making more outliers (values are same)

PINNs	wPINNs
 **Fig. 23** PINNs one (2150th) −1 manipulated	 **Fig. 24** wPINNs one (2150th) −1 manipulated
 **Fig. 25** PINNs three −1 manipulated	 **Fig. 26** wPINNs three −1 manipulated
 **Fig. 27** PINNs five −1 manipulated	 **Fig. 28** wPINNs five −1 manipulated

8. Reference

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wPINNs: 웨이블릿 활성화 함수를 활용한 물리 기반 신경망 향상

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물리 기반 신경망(PINNs)은 물리 법칙을 학습 프레임워크에 통합하여 편미분방정식 기반 문제를 해결하는 다재다능한 도구로 부상했습니다. 그러나 하이퍼볼릭 탄젠트 활성화 함수를 사용하는 MLP에 의존하는 기존 PINNs는 시간적 종속성을 모델링하는 데 종종 한계를 드러냅니다. 이러한 한계를 해결하기 위해 다중 헤드 주의 메커니즘과 웨이블릿 활성화 함수를 도입한 Transformer 기반 확장 모델인 PINNsFormer가 제안되었으나, 이는 상당한 계산 비용을 초래합니다. 본 연구는 웨이블릿 활성화 함수를 기존 PINNs 프레임워크에 통합한 wPINNs를 제안하며, 이는 높은 정확도를 제공하면서도 계산 비용을 크게 증가시키지 않습니다. 카르만 와류 문제를 대상으로 한 실험에서 wPINNs는 기존 PINNs에 비해 오차를 크게 낮추면서 계산 효율성을 유지하는 결과를 보였습니다. 또한, 이상치 데이터에 대한 강건성 테스트에서는 wPINNs가 합리적으로 이상치를 처리하는 반면, PINNs는 이상치 상황에서 예기치 못한 행동을 보이며 추가적인 조사가 필요함을 시사했습니다. 이러한 결과는 PINNs를 발전시키는 데 웨이블릿 활성화 함수가 유용함을 확인하며, 계산 물리학 문제를 해결하기 위한 실용적인 솔루션을 제공합니다.

주요어 : 웨이블릿 물리 기반 신경망, 물리 기반 신경망, 트랜스포머 기반 물리 신경망, 웨이블릿 활성화 함수, 나비에-스토크스 방정식, 카르만 와류

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